

Beniyam Demissie

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EDUCATION

University of Texas at San Antonio

August 2023 – May 2027

- Bachelor of Science in Computer Science (GPA: 3.1)
 - Relevant Courses: DSA, Discrete Math, Cyber Security, Systems & Application Programming, DB Systems

EXPERIENCE

IBM

May 2026 - August 2026

Software Developer Intern

- Shifted CVE remediation left with container security tooling covering 30+ vulnerability databases that diff's base vs. app image layers and auto-upgrades all fixable dependencies, cutting a multi-day scan-and-recall pipeline to 10 minutes.
- Built safe-fix and deep-fix AI agent skills resolving hundreds of CVEs in under 10 minutes across Node.js, Python, Java, and Go, including breaking major-version upgrades that take hours manually.
- Developed an agentic SonarQube workflow that scans, presents issues for developer selection, applies source-code fixes, and re-verifies, resolving hundreds of issues in 10 minutes vs. hours for 10 manually.
- Categorized detected CVEs by remediation status (auto-fixable, vendor patch pending, no fix available), turning raw scanner output into an immediate risk decision list.
- Shipped the toolkit as an MCP server and standalone agent skills; validated on 5 apps with 1,000+ dependencies and demoed to senior leadership, drawing interest from 4+ teams.

Texas Advanced Computing Center (TACC)

August 2025 – May 2026

Software Engineering Intern

Tacc Stats Team

December 2025 – May 2026

- Created a live jobs page (Django/Redis/SPA) with 15s auto-refresh, cutting per-incident triage from minutes to seconds.
- Designed a cluster heatmap visualizing hundreds of nodes with CPU color encoding by merging Redis and PostgreSQL data for at-a-glance researcher load detection.
- Added hover tooltips surfacing host, CPU, memory, jobs, and freshness per tile, eliminating UI hops during on-call triage.

Data Management & Collections Team

August 2025 – December 2025

- Deployed a research data catalog enabling 2,300+ researchers to publish, search, and share datasets through a single reliable portal on a Dockerized CKAN stack (PostgreSQL, Solr, Redis, Nginx).
- Configured the storage backend on Corral, TACC's high-performance storage system, leveraging 50+ PB with capacity to host several billion datasets and files without performance degradation.
- Integrated a TACC-branded frontend with 13 templates and 3 CSS bundles to support dataset discovery and entry.

OJO Labs

June 2022 – August 2022

Software Engineering Intern

- Optimized workflows for 1,000+ real estate agents by adding instant in-app loan status visibility, replacing 1–2 days of lender calls and record searches with real-time access that let agents focus on clients.
- Programmed 10+ React features integrated with Kotlin backend, extending the platform with new client profile views, loan management tools, and searchable data displays.
- Debugged 25+ commits in the CI/CD pipeline, resolving build failures and ensuring unit tests passed before merge.

PROJECTS

Repo Mind

May 2025 – August 2025

- Wrote a Retrieval-Augmented Generation system for codebases with index, upload, and query endpoints, enabling developers to search repositories conversationally; containerized with Docker + Gunicorn for reliable deployment.
- Cut query latency from ~1 min to 2–5s and hallucinations by ~27% via targeted ChromaDB code-chunk retrieval

TECHNICAL SKILLS

- **Languages & Web:** Python, Bash, Java, Kotlin, JavaScript/TypeScript, Node.js, SQL, HTML/CSS
- **Frameworks & Data:** FastAPI, Flask, React, PostgreSQL, SQLAlchemy, Solr, CKAN
- **AI & Agent Development:** Subagent orchestration, Custom agent skills, MCP servers, RAG, LLM embedding, ChromaDB
- **DevOps & Systems:** Docker, CI/CD, Gunicorn, Nginx, Linux/Unix, Git, Render